Distance-Sidon subsets from bounded-inertia number-field towers*

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Abstract

For a finite planar point set P, let $s(P)$ be the largest size of a subset whose pairwise distances are all distinct, and put

$$F_2(n) = \min_{|P|=n} s(P).$$

Recent work gives the window $n^{1/3} \ll F_2(n) \ll n^{1/2-\varepsilon}$ for some absolute $\varepsilon > 0$. We make the upper-bound saving explicit:

$$\boxed{F_2(n) \ll n^{0.49368323}}.$$

The construction combines three ingredients. We impose bounded tame inertia and bounded Frobenius order in a Golod–Shafarevich pro-2 tower; we place points in a product of disks in the Eisenstein CM extension of a totally real layer; and we amplify repeated algebraic norms using prime-power valuation patterns. A one-sided ideal-packing argument gives the geometric constant $2\sqrt{3}/\pi$. The tower is built over $\mathbb{Q}(\sqrt{11235917})$. Its class, ray, Frobenius, and numerical endpoint data are finite and checked by three executable implementations, including an independent reconstruction.

1 Introduction

Call a finite set $A \subset \mathbb{R}^2$ *distance-Sidon* if the map

$$\{x, y\} \longmapsto |x - y|, \qquad x, y \in A, \quad x \neq y,$$

is injective. For a finite planar set P, write

$$s(P) = \max\{|A| : A \subseteq P \text{ is distance-Sidon}\}, \qquad F_2(n) = \min_{|P|=n} s(P).$$

The problem of estimating $F_2(n)$ is recorded as Erdős problem 1208 [Blo]. Charalambides proved $F_2(n) \gg n^{1/3}/\log n$ [Cha13]; Clemen, Führer, and Roche-Newton recently removed the logarithm [CFRN26]. In the other direction, Lee, Pohoata, and Zhu constructed point sets for which every sufficiently large subset contains a repeated distance [LPZ26]. In particular, for some absolute $\varepsilon > 0$,

$$n^{1/3} \ll F_2(n) \ll n^{1/2-\varepsilon}. \tag{1.1}$$

Before this polynomial saving, Croot, Mao, Pohoata, Sheffer, and Yip obtained the first super-polylogarithmic improvement for distance-Sidon subsets of the square grid [CMP+26].

Our purpose is to give a compact quantitative version of the upper-bound construction.

*This investigation was carried out with substantial assistance from automated systems, and we omit a conventional byline rather than overstate the named researchers' personal contributions. *Nikol Savova*, University of Oxford, nikol.p.savova@gmail.com; *Sihao Huang*, independent researcher, sihao.c.huang@gmail.com.

Theorem 1.1. *For every positive integer n, there is an n-point set $P \subset \mathbb{R}^2$ such that every distance-Sidon subset of P has size $O(n^{0.49368323})$. Equivalently,*

$$F_2(n) \ll n^{0.49368323}.$$

The proof is an arithmetic amplification of the Minkowski-grid idea. Its architecture is easier to describe than the numerical exponent suggests. An infinite totally real pro-2 tower supplies fields of dyadic degree and bounded root discriminant. At many unramified primes, Frobenius has order at most two. After adjoining a primitive cube root of unity, those primes split, so a difference z admits several prescribed allocations of its valuations between the conjugate factors of $z\bar{z}$. If a subset had all distances distinct, these allocations could not occur too often. Comparing a lower bound from congruence classes with an upper bound for totally positive ideal elements yields the master inequality in Proposition 2.3.

Two refinements account for the explicit saving. First, forcing the square of each tame inertia generator to be trivial limits every ramification index to two while preserving the Frattini quotient. Second, a product of planar disks in an Eisenstein CM field has a better diameter-to-volume ratio than the coordinate boxes used in the initial construction. The remaining optimization is a one-dimensional concave knapsack problem.

The bounded-inertia input goes back to Hajir and Maire [HM01, HM02], building on Koch's class-tower criterion [Haj96, Koc02]; we use the later tower-cutting framework of Hajir, Maire, and Ramakrishna [HMR21]. The recent number-field construction for the unit-distance problem and Sawin's explicit version are direct methodological precedents [ABG+26, Saw26]. Näslund's subsequent ideal-lattice covolume refinement is the closest geometric analogue of our product-disk step [N26]. We state the tame, totally real Shafarevich presentation theorem in the precise form needed below. The field-specific calculations use PARI/GP [26]; the accompanying verifier reconstructs the relevant Kummer kernel, checks every selected prime, and evaluates the final inequalities at two independent precisions.

All logarithms below are natural.

2 The Eisenstein norm sieve

Let L be a totally real field of degree m, and put

$$K = L(\zeta_3), \qquad \zeta_3^2 + \zeta_3 + 1 = 0.$$

Choose one complex embedding of K above each real embedding of L. This embeds $\mathcal{O}_K$ as a lattice in $\mathbb{C}^m$.

2.1 A product of disks

For $R > 0$, let $\Delta_R \subset \mathbb{C}$ be a disk of area R^2, and set $\mathcal{D}_R = \Delta_R^m$. The complex Minkowski lattice has covolume

$$\operatorname{covol}(\mathcal{O}_K) = 2^{-m}\sqrt{|\operatorname{disc} K|}.$$

The relative discriminant of $L(\zeta_3)/L$ divides $3\mathcal{O}_L$, whence

$$\operatorname{covol}(\mathcal{O}_K) \le \left(\frac{\sqrt{3}}{2}\operatorname{rd}(L)\right)^m. \tag{2.1}$$

Averaging translates over a fundamental domain therefore proves the next elementary fact.

Lemma 2.1 (disk averaging)**.** *There is a translate of $\mathcal{D}_R$ containing at least*

$$\left(\frac{2R^2}{\sqrt{3}\,\mathrm{rd}(L)}\right)^m$$

elements of $\mathcal{O}_K$. If z is the difference of two elements in that translate and $\eta = z\bar{z} \in \mathcal{O}_L$, then, for every real embedding $\sigma : L \hookrightarrow \mathbb{R}$,

$$0 < \sigma(\eta) \le \frac{4}{\pi} R^2. \tag{2.2}$$

Proof. The first assertion is volume divided by covolume. A disk of area R^2 has diameter $2R/\sqrt{\pi}$. Under the complex embedding above σ, $\sigma(z\bar{z}) = |z|^2$, which proves (2.2). □

We shall use

$$R^2 = \frac{\sqrt{3}}{2}\,\mathrm{rd}(L) n^{1/m}. \tag{2.3}$$

Then the selected translate contains at least n lattice points. After projection through one distinguished complex embedding, it gives a planar point set. That projection is injective. Moreover, equality of two projected squared distances implies equality of the corresponding elements $z\bar{z} \in L$, because a field embedding is injective.

The following one-sided estimate is the reason that total positivity is useful.

Lemma 2.2 (one-sided ideal packing)**.** *Let $\mathfrak{a} \subset \mathcal{O}_L$ be a nonzero integral ideal and $Y > 0$. Then*

$$\#\{\eta \in \mathfrak{a} : 0 < \sigma(\eta) \le Y \text{ for every } \sigma : L \hookrightarrow \mathbb{R}\} \le \left(1 + \frac{Y}{\mathrm{N}(\mathfrak{a})^{1/m}}\right)^m. \tag{2.4}$$

Proof. Partition $(0, Y]^m$ into half-open cubes of side $\delta = \mathrm{N}(\mathfrak{a})^{1/m}$. Two embedded ideal elements in one cube would have a nonzero difference $\gamma \in \mathfrak{a}$ with $|\mathrm{N}_{L/\mathbb{Q}}\,\gamma| < \delta^m = \mathrm{N}\,\mathfrak{a}$. This is impossible because $(\gamma) \subseteq \mathfrak{a}$. There are at most $(1 + Y/\delta)^m$ cubes. □

2.2 Valuation patterns

Suppose a prime ideal $\mathfrak{q}$ of L splits in K/L as

$$\mathfrak{q}\mathcal{O}_K = \mathfrak{P}\bar{\mathfrak{P}}.$$

Fix a depth $k \ge 0$. For each $0 \le a \le k$, consider the additive lattice

$$I_a = \mathfrak{P}^a \bar{\mathfrak{P}}^{k-a} \subset \mathcal{O}_K. \tag{2.5}$$

It has index $\mathrm{N}(\mathfrak{q})^k$. Products over several prime ideals and possibly different depths give pattern lattices $I_{\mathbf{a}}$. Write

$$\mathfrak{M} = \prod_{\mathfrak{q}} \mathfrak{q}^{k_{\mathfrak{q}}}, \quad H = \prod_{\mathfrak{q}} (k_{\mathfrak{q}} + 1), \quad \Lambda = \prod_{\mathfrak{q}} \left(1 + \mathrm{N}\,\mathfrak{q}^{-1} + \cdots + \mathrm{N}\,\mathfrak{q}^{-k_{\mathfrak{q}}}\right), \tag{2.6}$$

and $\mu = \mathrm{N}(\mathfrak{M})^{1/m}$.

For a finite set $A \subset \mathcal{O}_K$, Cauchy–Schwarz over the cosets of every pattern lattice gives

$$\sum_{\mathbf{a}} \#\{(x, y) \in A^2 : x \ne y, x - y \in I_{\mathbf{a}}\} \ge H \frac{|A|(|A| - \mathrm{N}\,\mathfrak{M})}{\mathrm{N}\,\mathfrak{M}}. \tag{2.7}$$

On the other hand, if $z = x - y$ belongs to several patterns at a fixed prime, the excess multiplicity forces an extra power of $\mathfrak{q}$ to divide $z\bar{z}$. More precisely, the number of patterns containing z is at most the number of ideals $\mathfrak{b} \mid \mathfrak{M}$ for which

$$\mathfrak{M}\mathfrak{b} \mid (z\bar{z}). \tag{2.8}$$

This is the divisor switch: at $\mathfrak{P}$ and $\bar{\mathfrak{P}}$, the admissible values of a form the interval $[k - v_{\bar{\mathfrak{P}}}(z), v_{\mathfrak{P}}(z)] \cap [0, k]$.

If the planar projection of A is distance-Sidon, each nonzero value $z\bar{z}$ comes from at most two ordered pairs. Applying Lemma 2.2 to (2.8), and then summing $\mathrm{N}(\mathfrak{b})^{-1}$ over $\mathfrak{b} \mid \mathfrak{M}$, yields the master bound.

Proposition 2.3 (norm-sieve master inequality)**.** *Let A be a distance-Sidon subset of a translate of $\mathcal{D}_R$. With the notation above,*

$$|A| \le \mu^m + \sqrt{2} R^m \left[(\Lambda/H)^{1/m} \left(\frac{4}{\pi} + \frac{\mu^2}{R^2} \right) \right]^{m/2}. \tag{2.9}$$

Proof. By (2.2), every norm $z\bar{z}$ lies in the positive box with $Y = 4R^2/\pi$. For $\mathfrak{b} \mid \mathfrak{M}$, Lemma 2.2 gives

$$\#\{\eta \in \mathfrak{M}\mathfrak{b} : 0 < \sigma(\eta) \le Y\} \le \left[\frac{R^2}{\mu \, \mathrm{N}(\mathfrak{b})^{1/m}} \left(\frac{4}{\pi} + \frac{\mu^2}{R^2} \right) \right]^m,$$

where we used $\mathrm{N}(\mathfrak{b})^{1/m} \le \mu$. Sum over $\mathfrak{b} \mid \mathfrak{M}$, use $\sum_{\mathfrak{b} \mid \mathfrak{M}} \mathrm{N}(\mathfrak{b})^{-1} = \Lambda$, multiply by the ordered-pair factor two, and compare with (2.7). If $|A| \ge \mu^m$, then $(|A| - \mu^m)^2 \le |A|(|A| - \mu^m)$; taking square roots gives (2.9). The case $|A| < \mu^m$ is immediate. $\square$

Combining (2.3) with (2.9) replaces the real two-coordinate constant $4/\pi$ by the effective constant

$$C_{\mathrm{Eis}} = \frac{2\sqrt{3}}{\pi}. \tag{2.10}$$

3 A bounded-inertia pro-2 tower

We record the group-theoretic input in a form that separates the standard presentation theorem from the two elementary quotient operations used here.

For a finite set T of odd prime ideals of a number field E, put

$$V_T = \{a \in E^\times : (a) = \mathfrak{a}^2 \text{ for some fractional ideal } \mathfrak{a},\ a \in E_{\mathfrak{p}}^{\times 2} \text{ for every } \mathfrak{p} \in T\}/E^{\times 2}. \tag{3.1}$$

This is the usual tame Kummer obstruction group.

Theorem 3.1 (tame totally real Shafarevich–Koch presentation)**.** *Let E be totally real with r_1 real places, let T be a finite set of odd prime ideals, and let G_T^+ be the Galois group of the maximal pro-2 extension of E unramified away from T in which every real place splits. Then G_T^+ has a finite minimal presentation with generator and relation ranks satisfying*

$$d(G_T^+) = |T| - r_1 + \dim_{\mathbb{F}_2} V_T, \qquad r(G_T^+) \le |T| - 1 + \dim_{\mathbb{F}_2} V_T. \tag{3.2}$$

In particular, if E is real quadratic and $V_T = 0$, then

$$d(G_T^+) = |T| - 2, \qquad r(G_T^+) \le |T| - 1 = d(G_T^+) + 1. \tag{3.3}$$

Proof. This is the tame $p = 2$ presentation theorem of Koch [Koc02, Theorems 11.5 and 11.8]; see also [NSW08, Section 10.7]. In the standard formulas,

$$d = |T| - (r_1 + r_2) + 1 - \delta_2(E) + \dim_{\mathbb{F}_2} V_T, \qquad r \leq |T| - \delta_2(E) + \dim_{\mathbb{F}_2} V_T.$$

Here $r_2 = 0$ and $\delta_2(E) = 1$ because $-1 \in E$. The dyadic places lie outside T, while requiring all real places to split is the totally real local condition; this is the archimedean refinement that removes the extra $p = 2$ relation present for the unrestricted extension. Substitution gives (3.3). ☐

Proposition 3.2 (tower criterion)**.** *Let E be totally real, and let T be a finite set of odd prime ideals of E. Suppose the maximal totally real pro-2 extension unramified away from T has a presentation with generator rank d and relation rank at most r_0. Let U be a set of unramified prime ideals with chosen Frobenius elements. Form the quotient obtained by imposing*

$$\tau_{\mathfrak{p}}^2 = 1 \quad (\mathfrak{p} \in T), \qquad \mathrm{Frob}_{\mathfrak{q}}^2 = 1 \quad (\mathfrak{q} \in U), \tag{3.4}$$

where $\tau_{\mathfrak{p}}$ generates tame pro-2 inertia. If

$$4(r_0 + |T| + |U|) < d^2, \tag{3.5}$$

then this quotient is infinite. Its Frattini quotient is unchanged, and in every finite layer its ramification index at $\mathfrak{p} \in T$ is at most two.

Proof. Tame pro-2 inertia at an odd prime is procyclic. Normal closure of the single relation $\tau_{\mathfrak{p}}^2$ therefore caps all conjugate inertia groups above $\mathfrak{p}$; normal closure likewise makes the chosen Frobenius representative irrelevant. Every relation in (3.4) is a square and hence lies in the Frattini subgroup, so the generator rank remains d. Adding at most $|T| + |U|$ relations gives the relation bound in (3.5). The strict Golod–Shafarevich inequality then proves infinitude; see, for example, [Ers12]. The ramification assertion is immediate from the inertia cap. ☐

In the explicit field below, both $V_T = 0$ and the generator space are verified exactly. Thus Theorem 3.1, rather than a generic rank estimate, supplies the base relation bound.

At a tame prime $\mathfrak{p}$ of E, the relative different exponent in a finite layer is at most one half of the relative degree. Thus the tower in Proposition 3.2 has root discriminant bounded by

$$D_0 = \mathrm{rd}(E) \prod_{\mathfrak{p} \in T} \mathrm{N}_E(\mathfrak{p})^{1/(2[E:\mathbb{Q}])}. \tag{3.6}$$

For a real quadratic base, the exponent is $1/4$.

We also need the Frobenius caps to make primes split in the Eisenstein CM extension of each layer. Call an unramified prime ideal $\mathfrak{q}$ *useful* if it does not lie above 3 and, writing $Q = \mathrm{N}_E \, \mathfrak{q}$, either $Q \equiv 1 \pmod 3$, or $Q \equiv 2 \pmod 3$ and the Frobenius class of $\mathfrak{q}$ is nonzero in the Frattini quotient. In the second case it has exact order two after the square cap, and the residue field in the selected layers has size $Q^2 \equiv 1 \pmod 3$; in the first case splitting is automatic.

For completeness, an infinite finitely generated pro-2 group has, for every sufficiently large j, an open normal subgroup N_j contained in its Frattini subgroup and of index 2^j. Indeed, take arbitrarily large finite 2-group quotients below the Frattini subgroup and refine a normal series by central factors of order two. The corresponding Galois layers have every sufficiently large dyadic degree and preserve every nonzero Frattini class.

4 The local frontier

Let a prime ideal of the quadratic base have norm Q. Across an orbit in a tower layer, one extra unit of valuation depth has absolute-degree normalized cost

$$c_Q = \frac{1}{2}\log Q. \tag{4.1}$$

For $0 < t < 1$ and $k \geq 1$, put

$$A_k(t) = \frac{k+1}{k}\frac{1-t^k}{1-t^{k+1}}. \tag{4.2}$$

The guaranteed gain of the kth depth increment is

$$g_{Q,k} = \frac{1}{4}\log A_k(Q^{-2}). \tag{4.3}$$

This is exact when the residue degree is two and is a lower bound when it is one.

Lemma 4.1 (all-depth comparison)**.** *For $0 < t < 1$ and $k \geq 1$,*

$$A_k(t)^2 \geq A_k(t^2).$$

Moreover $A_k(t)$ decreases with k.

Proof. After cancelling positive factors, the first assertion is equivalent to

$$(1-t)\left(\sum_{j=0}^{2k} t^j - (2k+1)t^k\right) \geq 0,$$

which is AM–GM. For the second, write

$$b_k(t) = \frac{1+t+\cdots+t^k}{k+1} = \frac{1}{1-t}\int_t^1 x^k\,dx.$$

Cauchy–Schwarz makes the moment sequence $b_k(t)$ log-convex, while $A_k(t) = b_{k-1}(t)/b_k(t)$. □

Sort all available items $(c_Q, g_{Q,k})$ by decreasing slope $g_{Q,k}/c_Q$, subject to taking the depths at each prime in order. Let $F(x)$ be the fractional-knapsack gain at total cost x. By Lemma 4.1, F is increasing, concave, and realizable in high degree up to an error $O(1/m)$: assign the final fractional item to the largest collection of prime ideals whose total residue degree remains below the desired fraction.

Put

$$w = \frac{\log n}{2m}, \qquad x = \log \mu.$$

Set $x = 2\alpha w$, so the first term in (2.9) is n^α. Using (2.3), the second term is at most a constant times n^α provided

$$F(2\alpha w) \geq \log(C_{\mathrm{Eis}}D_0) + (2-4\alpha)w + \log\left(1 + \frac{e^{2(2\alpha-1)w}}{C_{\mathrm{Eis}}D_0}\right). \tag{4.4}$$

The difference between the two sides is concave in w: the first term is concave, the middle term is affine, and $-\log(1+ae^{-cw})$ is concave for $a, c > 0$. Consequently it is enough to check (4.4) at the two endpoints of an interval $[w_0, 2w_0]$.

Lemma 4.2 (phase criterion)**.** *Suppose the tower has layers of every sufficiently large dyadic degree and* (4.4) *holds with a fixed positive margin at* $w = w_0$ *and* $w = 2w_0$*. Then* $F_2(n) \ll n^\alpha$*.*

Proof. For every sufficiently large n, choose a tower layer of degree m such that $w = (\log n)/(2m)$ lies in $[w_0, 2w_0)$. Concavity gives (4.4); the fixed endpoint margin absorbs the $O(1/m)$ placewise rounding error. Apply Proposition 2.3 to the n ambient points supplied by Lemma 2.1. The finitely many smaller values of n are absorbed by the implied constant. □

5 The explicit tower and certificate

We now specialize to

$$E = \mathbb{Q}(\sqrt{D}), \qquad D = 11235917 = 7 \cdot 11 \cdot 337 \cdot 433. \tag{5.1}$$

The ring of integers has basis $1, \omega$, where

$$\omega^2 - \omega - 2808979 = 0.$$

Let T be the first 217 odd prime ideals of E, ordered first by norm and then by the underlying rational prime; ties between the two split ideals above one rational prime are broken by the increasing residue of ω.

Proposition 5.1 (finite arithmetic certificate)**.** *For the field and prime set above, the following statements hold.*

1. $\mathrm{Cl}(E) \cong C_{14} \times C_2$ *and* $\mathrm{Cl}^+(E) \cong C_{14} \times C_2 \times C_2$*. The* T*-class group is trivial.*
2. *The space of squareclasses* $aE^{\times 2}$ *with* (a) *an ideal square has dimension four. Its* 4×217 *matrix of local quadratic characters at* T *has rank four. Consequently* $V_T = 0$*.*
3. *The* 219 *fundamental* T*-unit squareclasses have rank-four image under the two sign and modulo-*4 *conditions. The safe Kummer kernel therefore has dimension* $d = 215$*.*
4. *There are* 11123 *useful prime ideals after* T*, with no rejection before the cutoff. The last selected and useful ideals have data*
$$(1063, 1063, 963), \qquad (121951, 121951, 70091), \tag{5.2}$$
where the triples record norm, underlying rational prime, and the residue of ω*.*
5. *The tower quotient has relation rank at most*
$$r = 216 + 217 + 11123 = 11556, \qquad 4r = 46224 = 215^2 - 1 < 215^2. \tag{5.3}$$
Its root discriminant satisfies
$$\log D_0 = \frac{1}{2}\log D + \frac{1}{4}\sum_{\mathfrak{p}\in T} \log \mathrm{N}\,\mathfrak{p} = 318.95275859164604144277651 65\ldots. \tag{5.4}$$

The class and ray assertions in Proposition 5.1 are certified by PARI/GP with `bnfcertify`=1, so they are unconditional and use no GRH assumption. They are not used as opaque numerical output: the verifier expands the returned T-units in the basis $1, \omega$, recomputes both real signs and the square classes modulo 4, and independently obtains the same rank-four row space. It also constructs the four global ideal-square classes and verifies rank four of their local-character matrix, proving $V_T = 0$. For every candidate norm congruent to two modulo three, it evaluates the quadratic-character functional on this exact 215-dimensional kernel. All 11123 required ideals pass.

The numerical part uses the rigorous rational estimate

$$C_{\mathrm{Eis}} = \frac{2\sqrt{3}}{\pi} < \frac{71603}{64935}. \tag{5.5}$$

It constructs the sorted frontier from (4.1)–(4.3). At

$$\alpha = 0.49368323, \qquad w_0 = 39928.00429278135001166535 4078\ldots, \tag{5.6}$$

the margins in (4.4) at w_0 and $2w_0$ are respectively

$$0.0016328170368155722502190 59\ldots, \qquad 0.0032470752120377069950298 33\ldots. \tag{5.7}$$

The active terminal slopes are approximately 0.0306672 and 0.0191268. The largest omitted fourth-depth slope is

$$0.0099999734338455349385051 68\ldots, \tag{5.8}$$

so Lemma 4.1 excludes every omitted later depth. The two fixed-anchor threshold crossings lie below 0.49368323; equalizing them gives

$$\alpha_* = 0.4936832199308881880151091959\ldots. \tag{5.9}$$

Thus the advertised exponent clears the equalized threshold by only $1.0069\ldots \times 10^{-8}$. The relation budget in (5.3) is likewise deliberately saturated at the largest integer allowed by the strict quadratic Golod–Shafarevich inequality: $4r = d^2 - 1$.

As insurance against weakening the presentation interface, the verifier also recomputes desaturated certificates. If one assumes only $r_0 \le d + 2$, omitting the final useful prime gives threshold $0.4936833032744\ldots$ and hence the safe bound $F_2(n) \ll n^{0.4936834}$. Under $r_0 \le d + 6$, omitting the final five useful primes gives threshold $0.4936836367746\ldots$ and the safe bound $F_2(n) \ll n^{0.4936837}$.

Proof of Theorem 1.1. By Proposition 5.1 and the tame Shafarevich bound (3.3), the strict inequality (5.3) allows Proposition 3.2 to produce an infinite totally real tower with root discriminant at most D_0 and all selected useful primes of residue degree at most two. The Frobenius test ensures that those primes split after passing to the Eisenstein CM extension. Equations (5.5)–(5.8) verify the two hypotheses of Lemma 4.2. The lemma gives $F_2(n) \ll n^{0.49368323}$. □

6 Verification and scope

The proof has a conceptual part and a finite certificate. The conceptual part is contained in the lemmas and propositions of Sections 2–4; none of those statements depends on the particular quadratic field. The certificate checks:

- the BNF, ordinary and narrow class groups, and localized class group;
- the four-dimensional ideal-square space and the equality $V_T = 0$;
- the exact T-unit basis and its sign/modulo-4 kernel;

- every selected prime ideal and every Frobenius usefulness test;
- the integer Golod–Shafarevich inequality, both desaturated fallbacks, and the root discriminant;
- the complete active depth frontier, omitted-depth cutoff, and both endpoint inequalities at 100 and 150 decimal digits.

Each endpoint computation is stable on increasing the working precision from 100 to 150 decimal digits, while the advertised margins exceed 10^{-3}. The repository includes a second hostile verifier which independently reconstructs the Kummer row space and repeats the arithmetic and endpoint checks; its output agrees far beyond the displayed precision.

Data and code availability. The source package accompanying this manuscript contains a dependency-closed, single-field verifier, its expected output, and the independent hostile reconstruction. The public repository is `https://github.com/NikolSavova/proof_hunter`. The immutable certificate snapshot is the repository at commit `c6f68c4`.

This paper does not determine the order of $F_2(n)$. The gap between $1/3$ and 0.49368323 remains substantial. Nor do we claim that $\mathbb{Q}(\sqrt{11235917})$ is globally optimal for the construction. The point of the explicit field is to make the quantitative theorem finite and reproducible. A significant approach toward the conjectural cube-root scale will require a new global mechanism rather than further decimal optimization of the same tower.

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